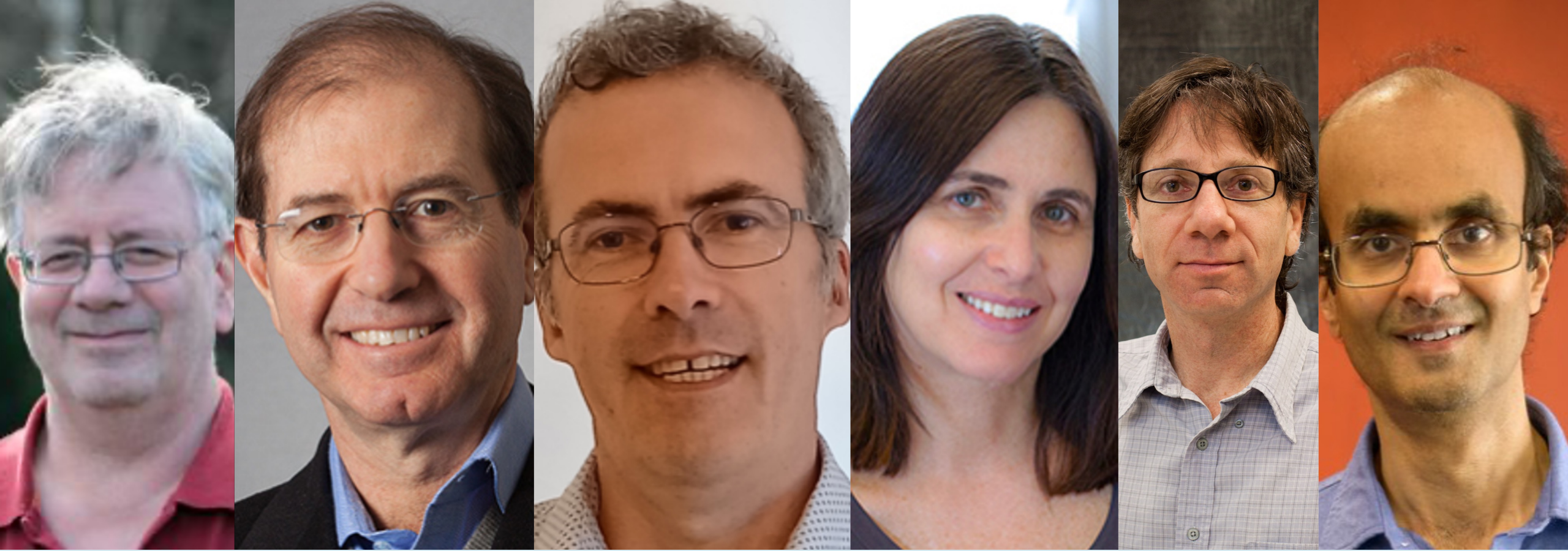


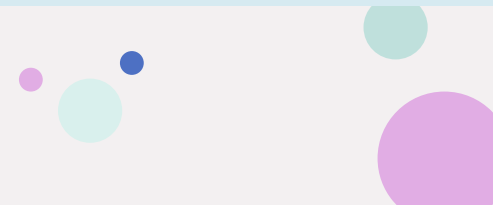


Lecture 6

Reusing same key, PRFs

CS409: Introduction to Cryptography

Instructor: Sruthi Sekar



Recap

- PRGs – unpredictability and pseudorandomness
- PRG Length Extension
- Computational Indistinguishability against Eavesdroppers
(only single message encrypted)
- Proof by Reduction
- Hybrid Arguments

Today's Class



Pseudorandom Functions (PRFs)



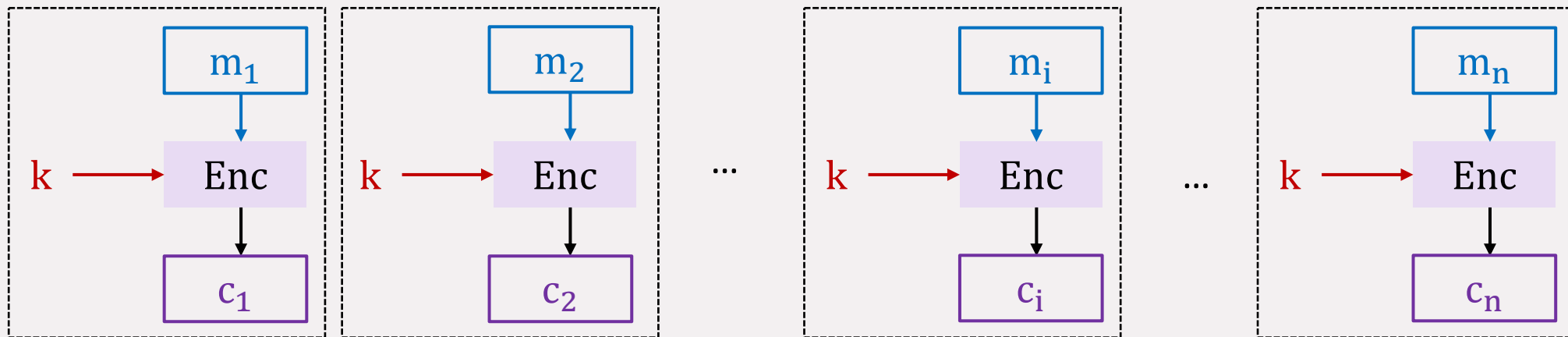
Encrypting multiple messages
with the same key

Encrypt many times with same key

$\Pi = (\text{Gen}, \text{Enc}, \text{Dec})$ be a computationally secure encryption scheme for ℓ -bit messages.

$k \leftarrow \text{Gen}$

GOAL: Encrypt multiple ℓ -bit messages (say n messages) using the same key k

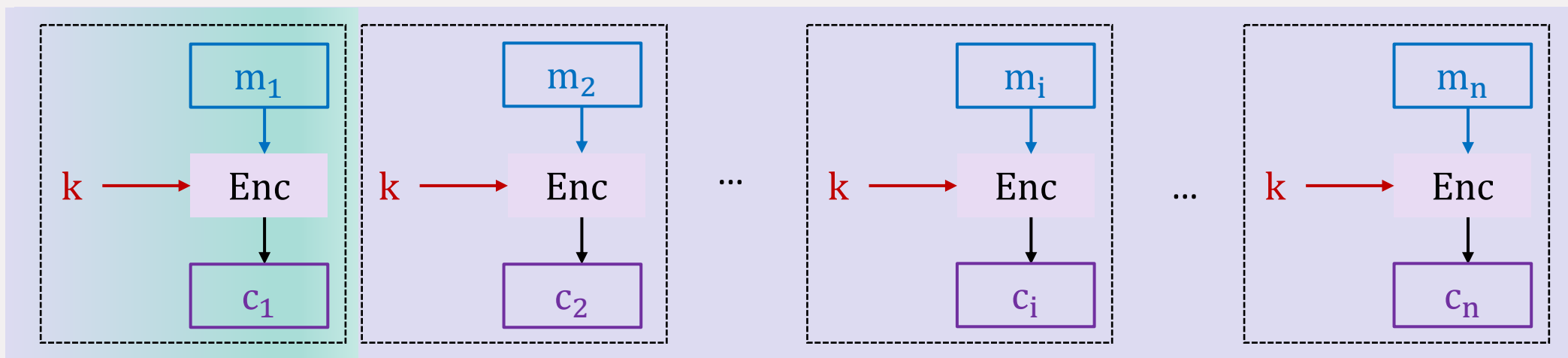


Encrypt many times with same key

$\Pi = (\text{Gen}, \text{Enc}, \text{Dec})$ be a computationally secure encryption scheme for ℓ -bit messages.

$k \leftarrow \text{Gen}$

GOAL: Encrypt multiple ℓ -bit messages (say n messages) using the same key k



Security of Multiple Encryptions:

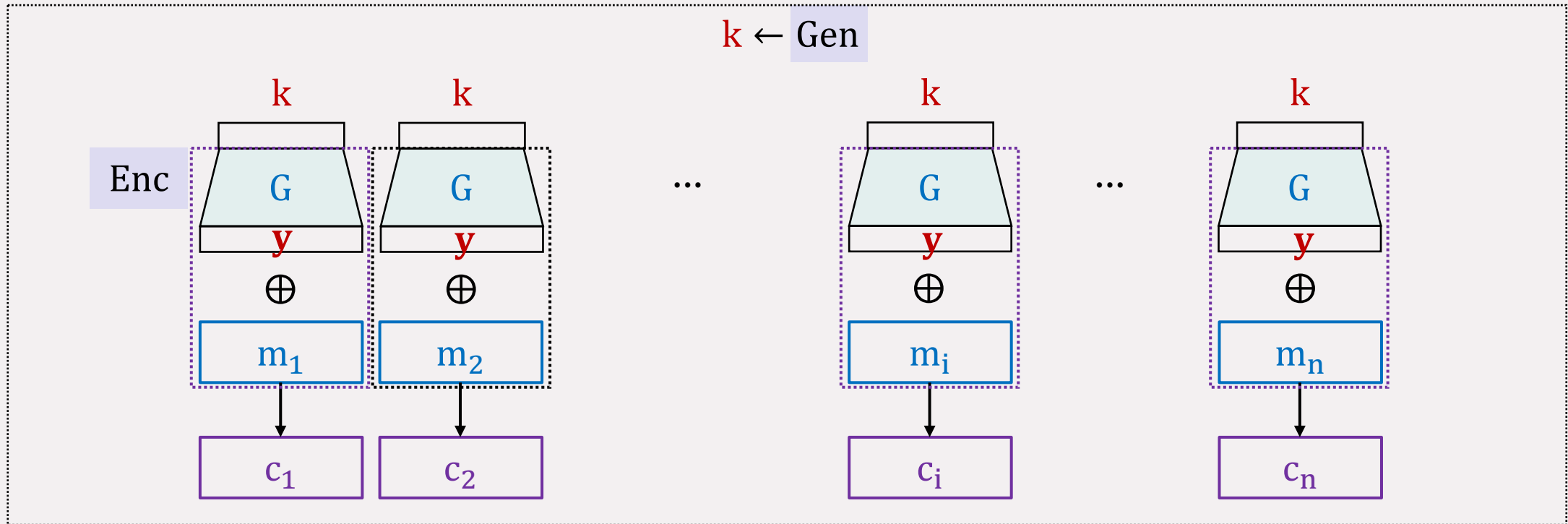


If Π is computationally secure, then the above scheme, $\Pi^{\otimes n}$, is computationally secure.

Is this true in general?

Attempt I

$\Pi = (\text{Gen}, \text{Enc}, \text{Dec})$ be the computationally one time pad for ℓ -bit messages. $G: \{0,1\}^n \rightarrow \{0,1\}^\ell$ be a PRG



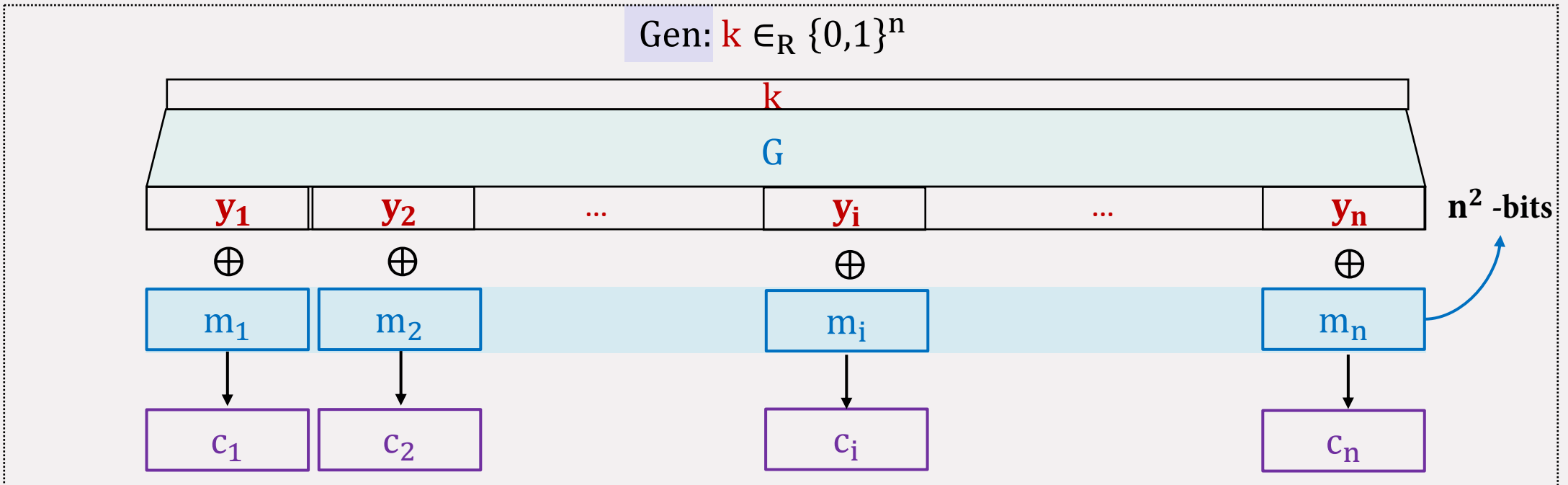
Computational One Time Pad for multiple encryptions

**Not
Secure**

As Eve, take messages $M = (m_1, m_2)$ and $M' = (m_1, m_1)$ (assume $n = 2$)

Attempt II

$G: \{0,1\}^n \rightarrow \{0,1\}^{n^2}$ be a PRG

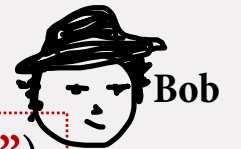
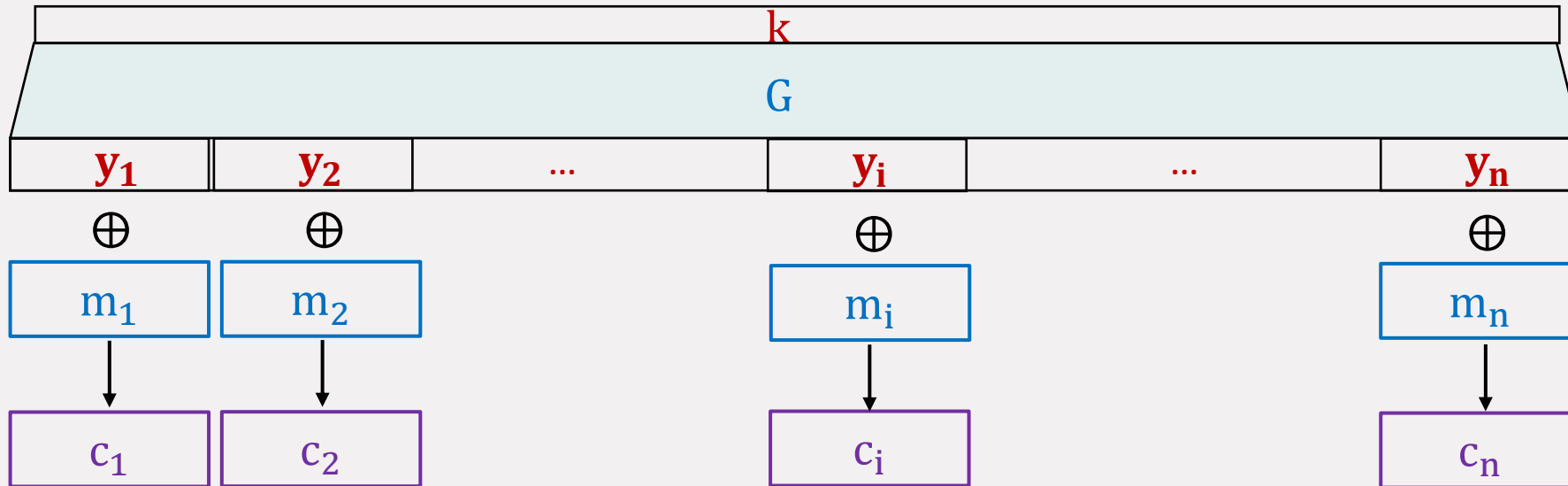


Variant of Computational One Time Pad for encrypting n messages, each n -bits long

Attempt II

$G: \{0,1\}^n \rightarrow \{0,1\}^{n^2}$ be a PRG

Gen: $k \in_R \{0,1\}^n$



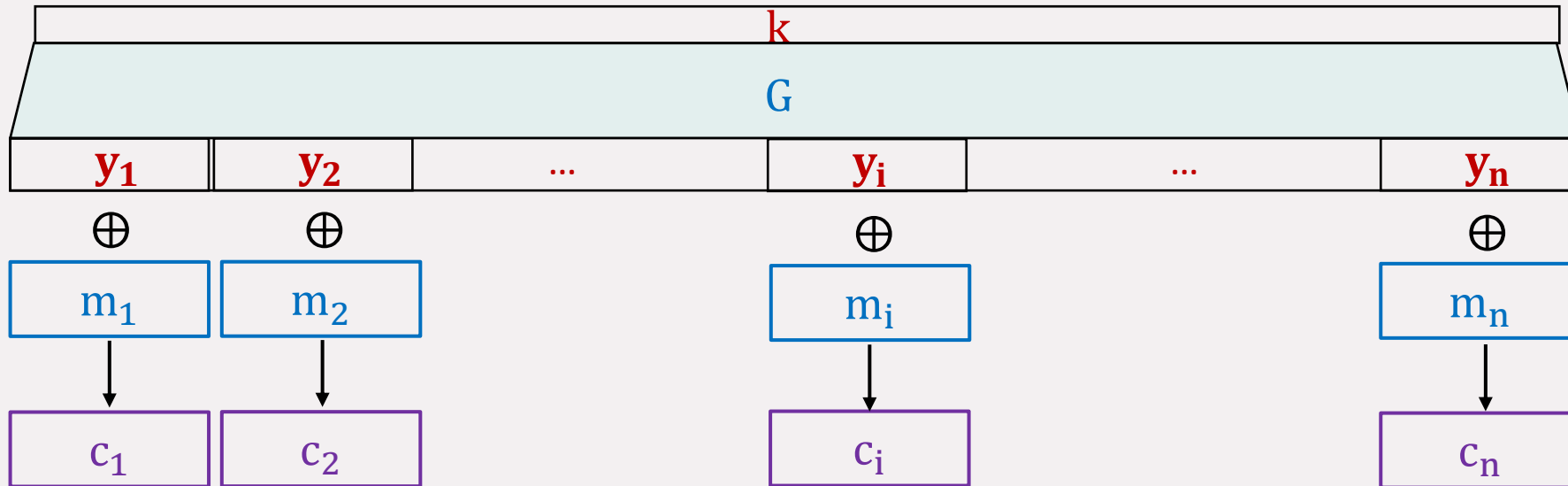
Alice and Bob need to know order of messages sent/received (i.e., need to be “stateful”)

Secure (and correct), if and only if Alice and Bob are stateful

Attempt II

$G: \{0,1\}^n \rightarrow \{0,1\}^{n^2}$ be a PRG

Gen: $k \in_R \{0,1\}^n$



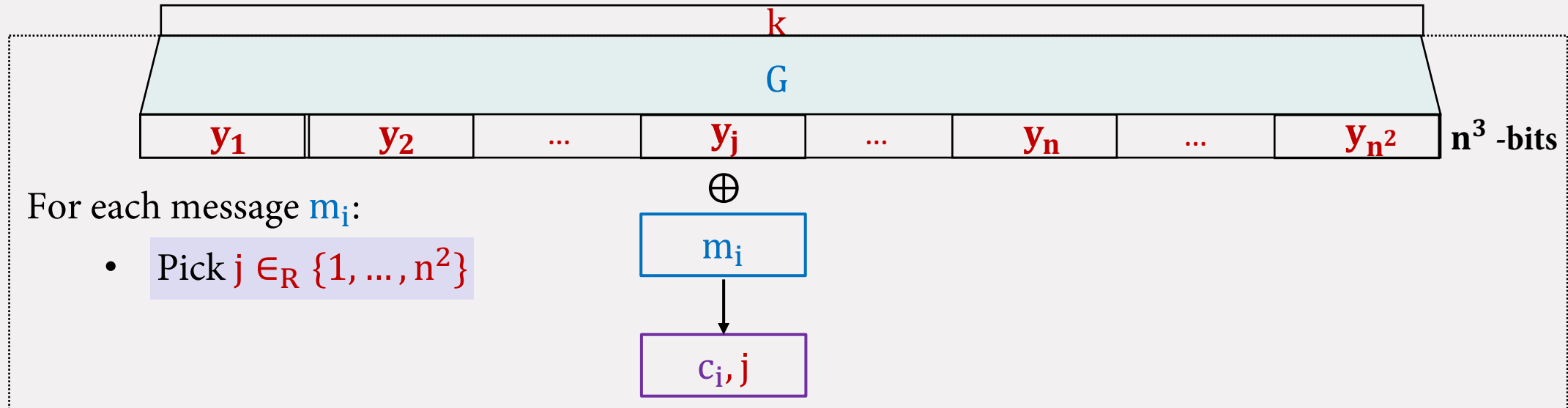
HW 1: Can you show the insecurity of multiple encryptions for any (stateless) deterministic encryption?

 [Katz-Lindell]

Attempt III

$G: \{0,1\}^n \rightarrow \{0,1\}^{n^3}$ be a PRG

Gen: $k \in_R \{0,1\}^n$



Randomized Computational One Time Pad: encrypt n messages each of length n -bits

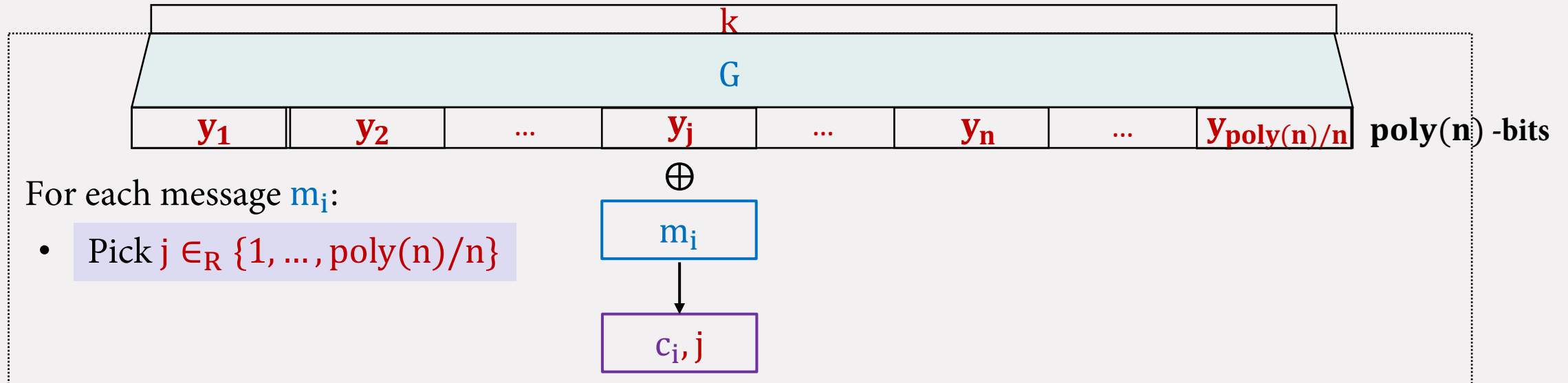


How often will you get collisions? (2 indices collide $\geq 1/n^2$ probability) **✗ NOT NEGLIGIBLE!**

Attempt III

$G: \{0,1\}^n \rightarrow \{0,1\}^{\text{poly}(n)}$ be a PRG

Gen: $k \in_R \{0,1\}^n$



Randomized Computational One Time Pad: encrypt n messages each of length n -bits



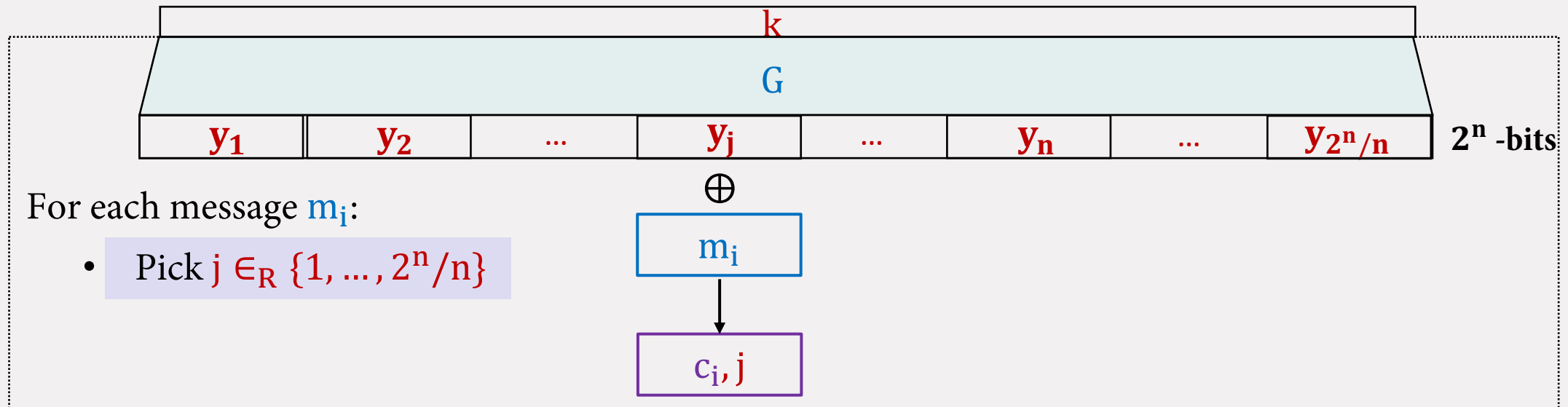
How often will you get collisions?

In general, same problem even with poly-stretch (collision with high probability).

Attempt IV

$G: \{0,1\}^n \rightarrow \{0,1\}^{2^n}$ be a PRG **Exponential stretch**

Gen: $k \in_R \{0,1\}^n$



Randomized Computational One Time Pad: encrypt $\text{poly}(n)$ messages each of length n -bits



How often will you get collisions? $\Pr[\exists \text{ collision in } t = \text{poly}(n) \text{ indices}] \leq t^2 n / 2^{n+1} = \text{negl}(n)$



Encryption and Decryption are not $\text{poly}(n)$ -time anymore ☹



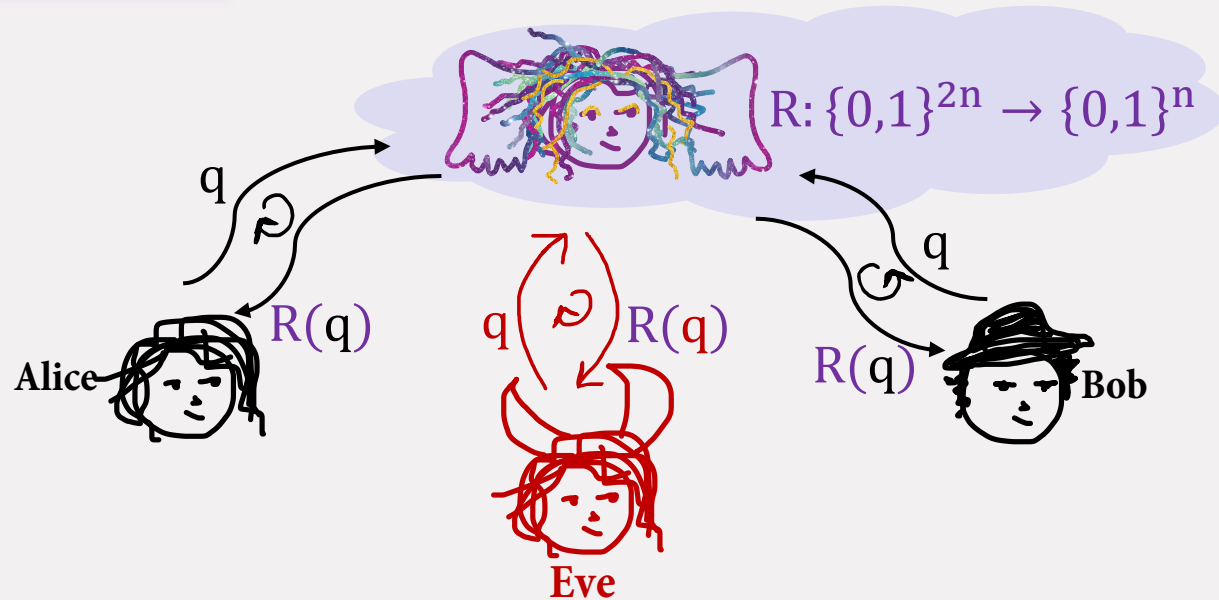
Birthday Bound: [Appendix A, Katz-Lindell]



Need: exponential one time pads, that are efficiently accessible

Attempt V: with some powers

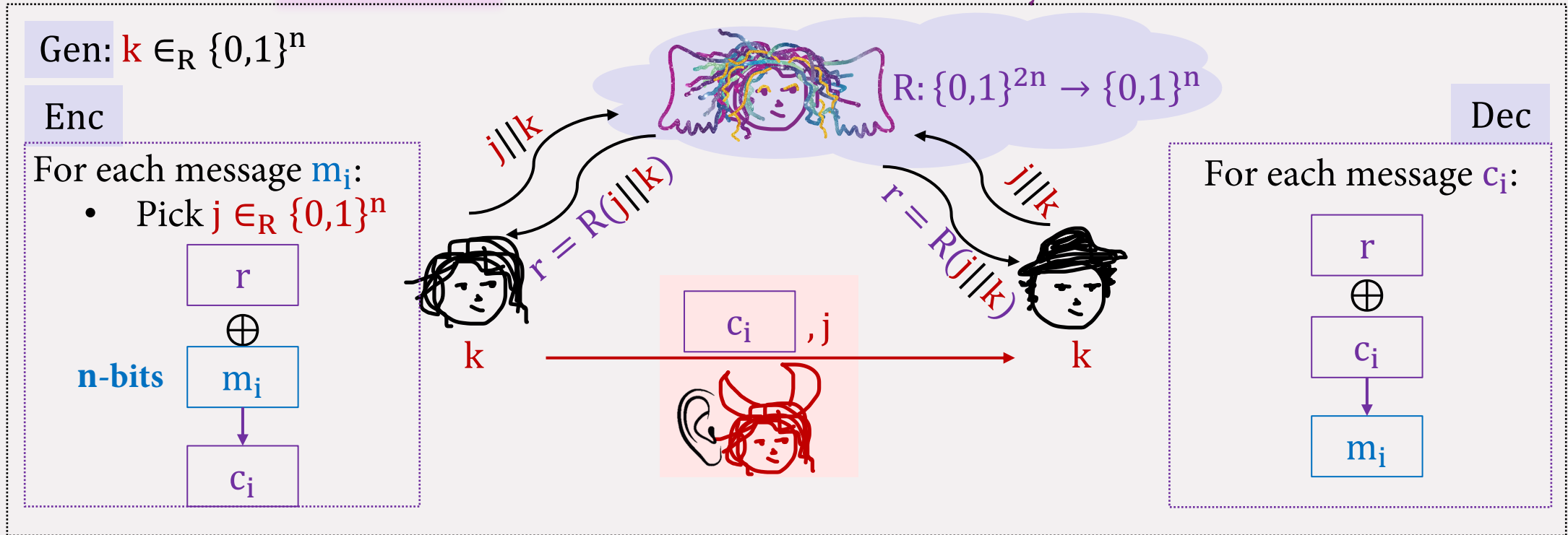
THE POWER: Random Function Oracle in the Sky accessible to all



Everyone (including **Eve**) has access to a random function oracle $R: \{0,1\}^{2n} \rightarrow \{0,1\}^n$

Attempt V: with some powers

THE POWER: Random Function Oracle in the Sky accessible to all



Computational One Time Pad with Random Oracle



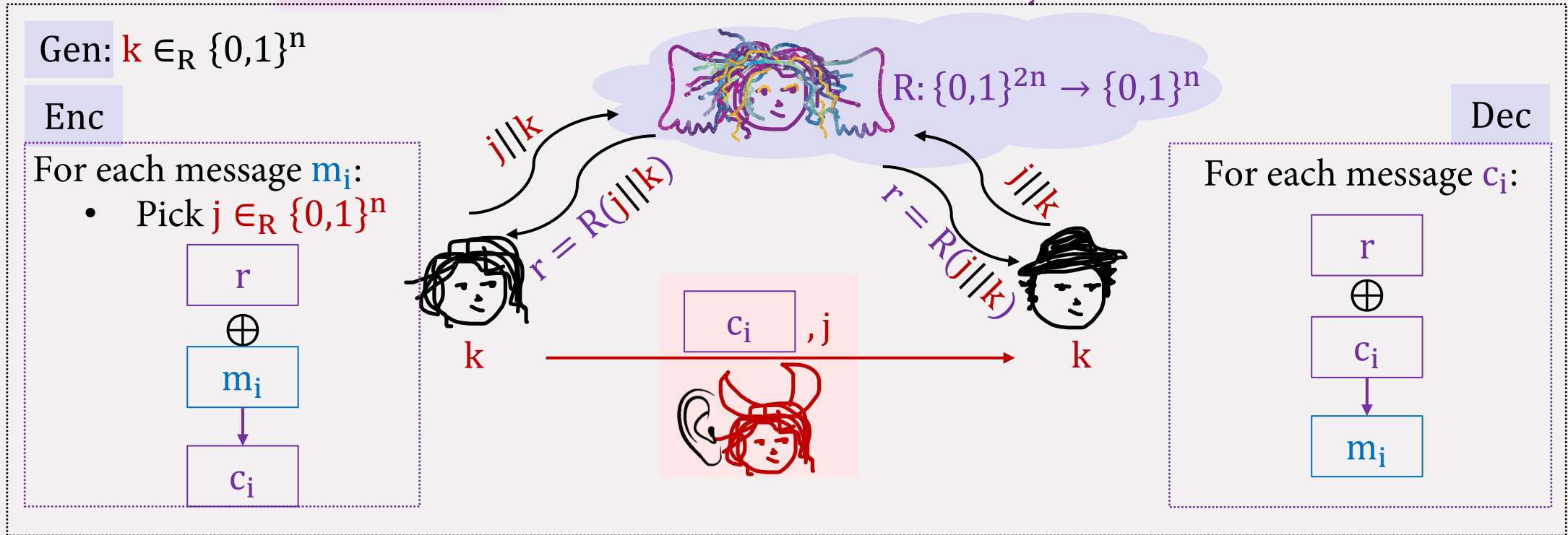
How will you use this R to fix the issue with Attempt IV and build a “stateless” encryption?



(R helps you generate exponentially many random OTPs!)

Attempt V: with some powers

THE POWER: Random Function Oracle in the Sky accessible to all



Computational One Time Pad with Random Oracle

HW 2 (Thought Exercise): Can Alice and Bob communicate securely if they did not have the shared key k but had access to this Random Oracle?

Today's Class



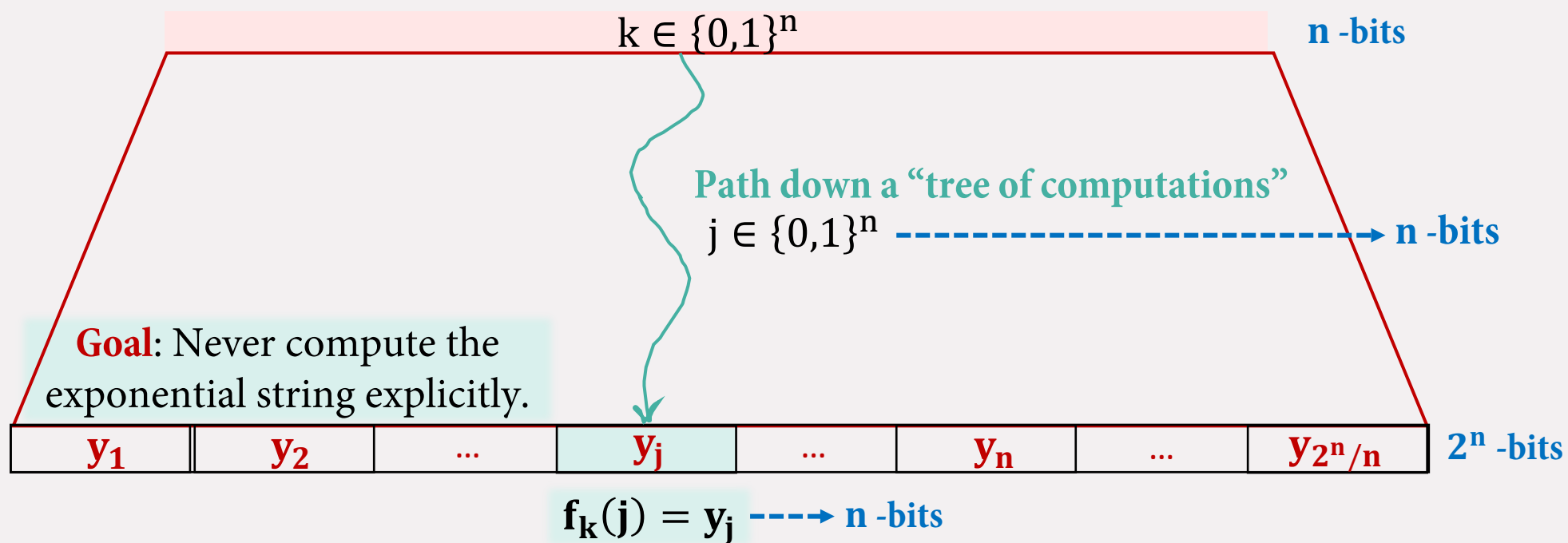
Pseudorandom Functions (PRFs)

Computational Analogue of Random Oracle



Encrypting multiple messages
with the same key

Motivating Pseudorandom Functions



We want some function f_k that computes $f_k(j) = y_j$, the “pseudorandom” string we want for OTP.

in **poly**($|x|$) = **poly**(n) time!

$f_k(j_1), f_k(j_2), \dots$ is computationally indistinguishable from random $r_1, r_2, \dots$

Seems like a random function oracle to PPT Eve

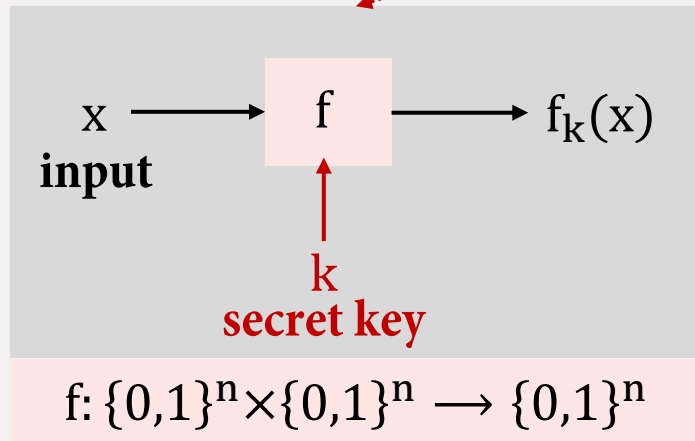


Need: exponential “pseudorandom” one time pads, that are efficiently accessible

Defining Pseudorandom Functions



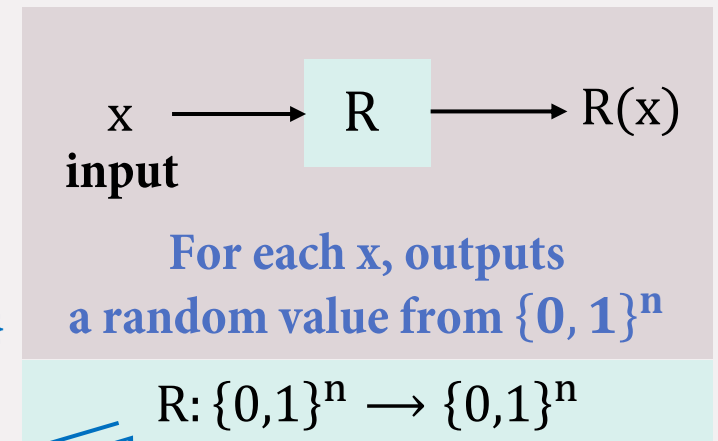
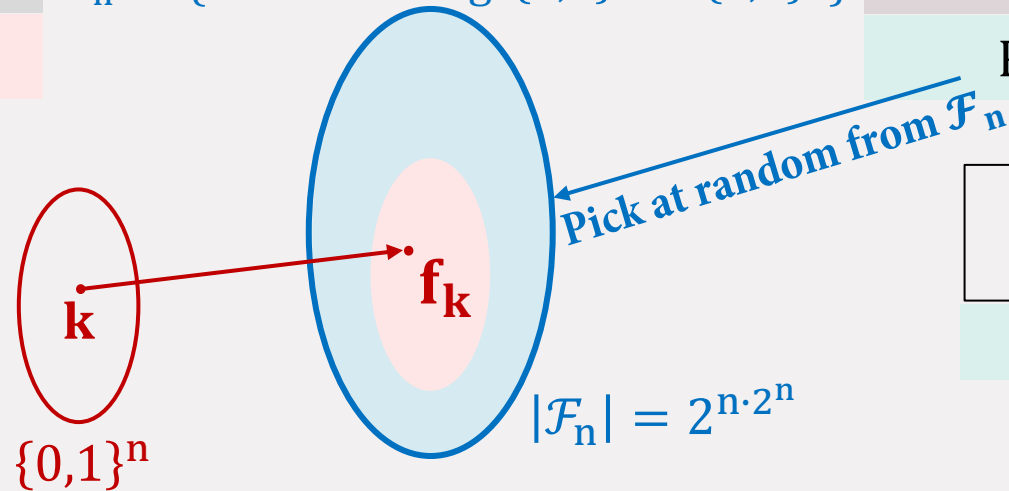
Want to define a function (taking a secret key) that “seems like” a random function oracle to **PPT** Eve.



Chosen from how many distinct functions?

2^n

$\mathcal{F}_n := \{\text{all functions } g: \{0,1\}^n \rightarrow \{0,1\}^n\}$



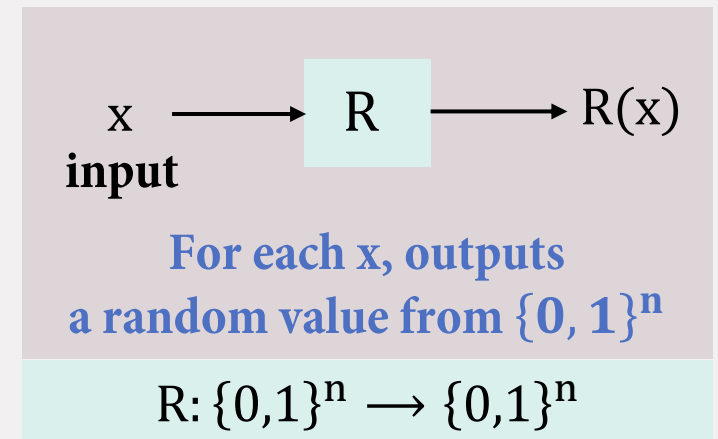
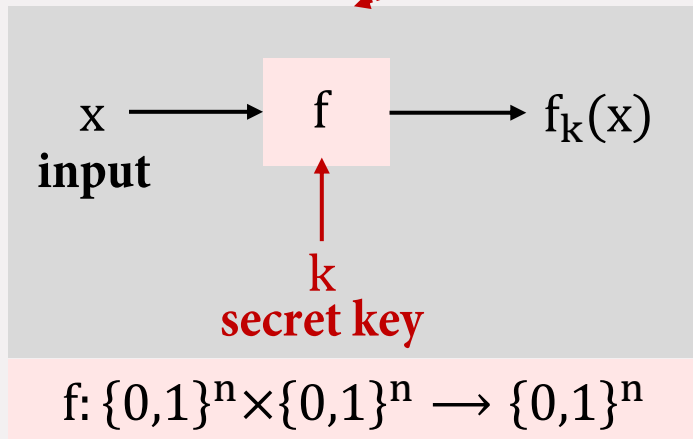
Chosen from how many distinct functions?

$2^{n \cdot 2^n}$

Defining Pseudorandom Functions



Want to define a function (taking a secret key) that “seems like” a random function oracle to **PPT** Eve.



RECALL: PRGs were defined as

**Pseudorandom
world**

$k \in_R \{0,1\}^n$
Output: $y = G(k)$

$\approx_C$
Computationally
indistinguishable

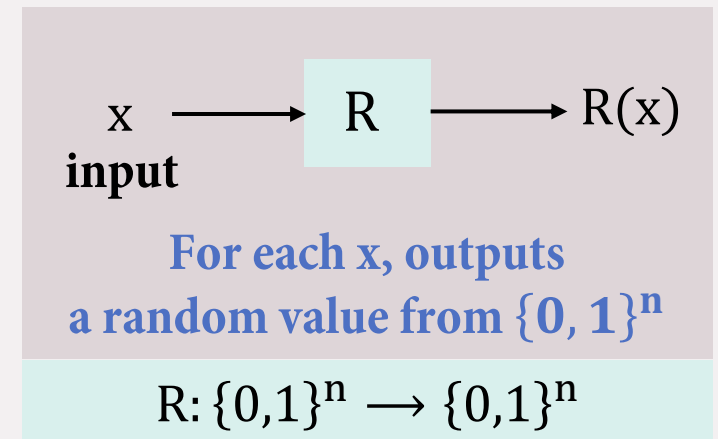
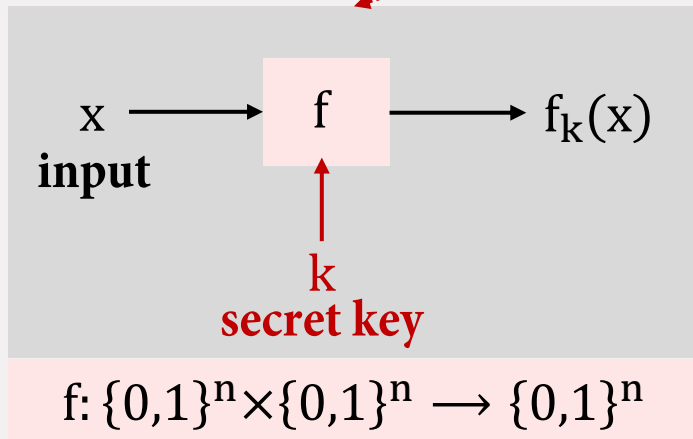
$y \in_R \{0,1\}^{\ell(n)}$
Output: y

**Random
world**

Defining Pseudorandom Functions



Want to define a function (taking a secret key) that “seems like” a random function oracle to **PPT** Eve.



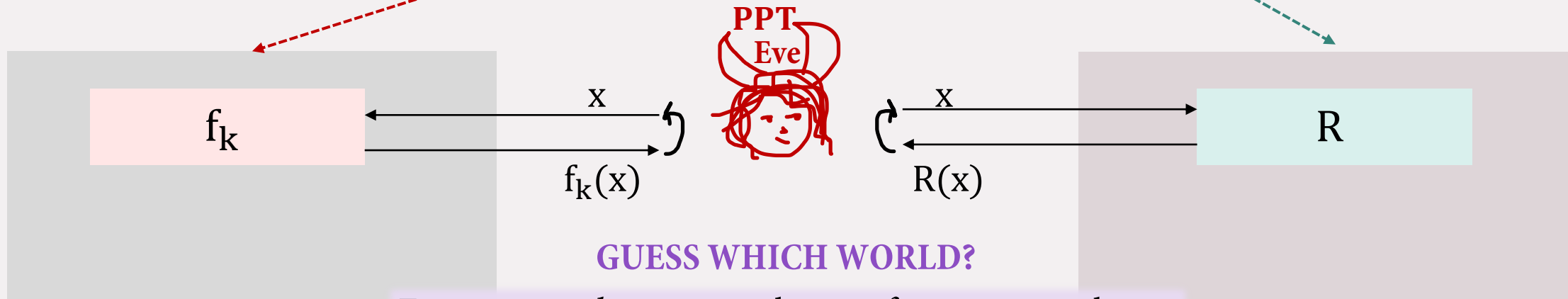
Can we give Eve the full description of either function (as a table, say)?

NO

Defining Pseudorandom Functions



Want to define a function (taking a secret key) that “seems like” a random function oracle to **PPT** Eve.



Eve gets *oracle access* to the two functions, with one query counted as one unit of running time

Definition 1

A family of functions $\{f_k: \{0,1\}^n \rightarrow \{0,1\}^n\}_{k \in \{0,1\}^n}$ is a **pseudorandom function** if for every **PPT** Eve, there exists a negligible function **negl** such that, $\forall n$

$$\left| \Pr_{k \in_R \{0,1\}^n} [\text{Eve}^{f_k(\cdot)}(1^n) = 1] - \Pr_{R \in_R \mathcal{F}_n} [\text{Eve}^{R(\cdot)}(1^n) = 1] \right| \leq \text{negl}(n)$$

Defining Pseudorandom Functions

HW 3: Prove
Equivalence of
Def 1 and 2.

Challenger



$$k \in_R \{0,1\}^n$$

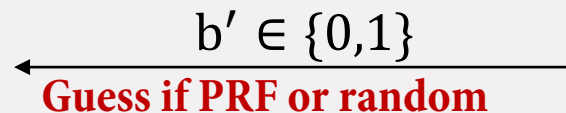
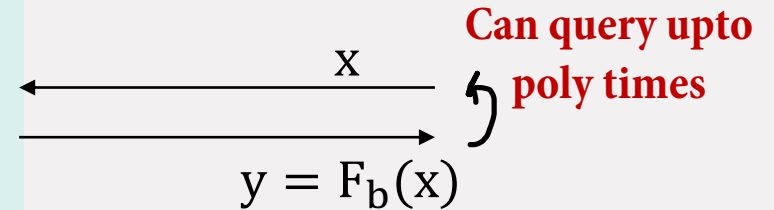
$$R \in_R \{\text{all functions } g: \{0,1\}^n \rightarrow \{0,1\}^n\}$$

Let $F_0 = f_k$

$$F_1 = R$$

Pick $b \in_R \{0,1\}$

$$\mathcal{F} = \{f_k: \{0,1\}^n \rightarrow \{0,1\}^n\}_{k \in \{0,1\}^n}$$



Eve wins if output is 1
 $\text{Dist}_{\text{Eve}, \mathcal{F}}$ outputs 1, if $b' = b$
and 0, o.w.

Definition 2

A family of functions $\{f_k: \{0,1\}^n \rightarrow \{0,1\}^n\}_{k \in \{0,1\}^n}$ is a **pseudorandom function** if for every **PPT Eve**, there exists a negligible function **negl** such that, $\forall n$

$$\Pr[\text{Dist}_{\text{Eve}, \mathcal{F}}(n) = 1] \leq 1/2 + \text{negl}(n)$$

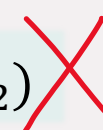
Quiz Time

PRF or not? ($F^{(1)}, F^{(2)}$ are PRFs)

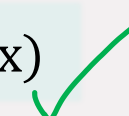
$$F_k(x) = k \oplus x$$



$$F_k(x_1 x_2) = F_k^{(1)}(x_1) F_k^{(2)}(x_2)$$



$$F_{k_1 k_2}(x) = F_{k_1}^{(1)}(x) F_{k_2}^{(2)}(x)$$



HW 4: Prove this formally.

PRG or not? (F is a PRF)

$$G(s) = F_s(1) F_s(2) \dots F_s(n-1) F_s(n)$$



$$G(s) = F_s(2^0) F_s(2^1) \dots F_s(2^{n-1}) F_s(2^n)$$

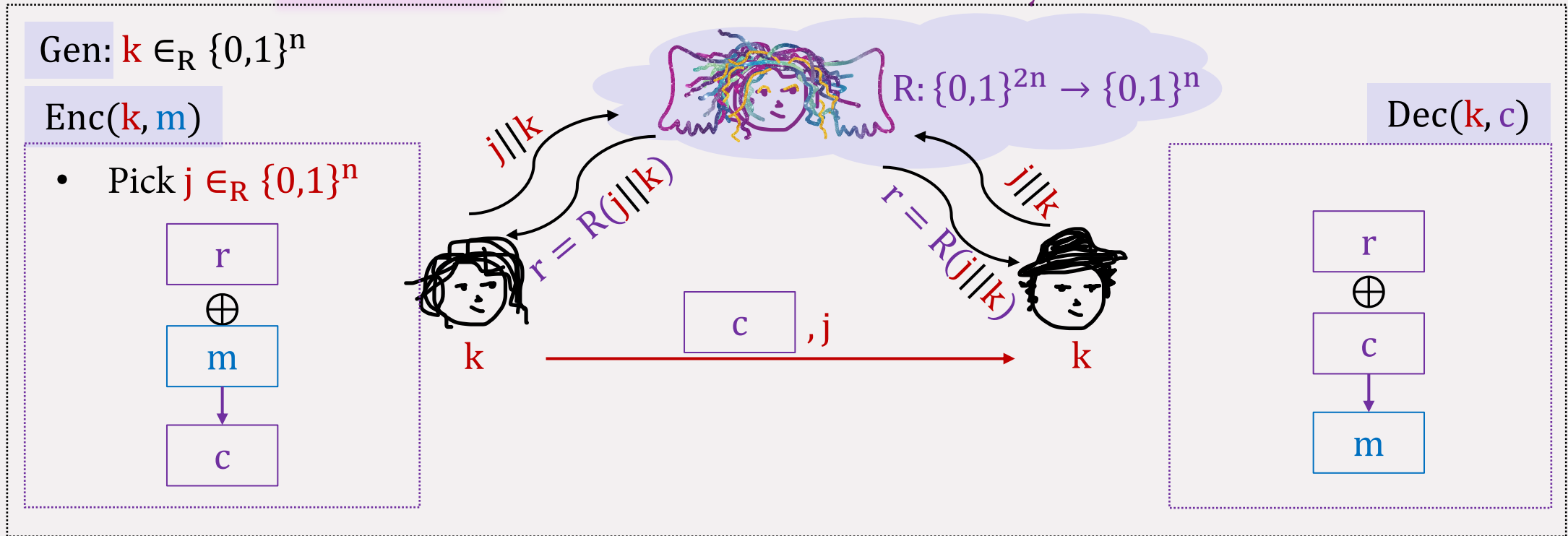


$$G(s) = F_1(s) F_2(s) \dots F_{n-1}(s) F_n(s)$$



Stateless Symmetric Key Encryption

THE POWER: Random Function Oracle in the Sky accessible to all

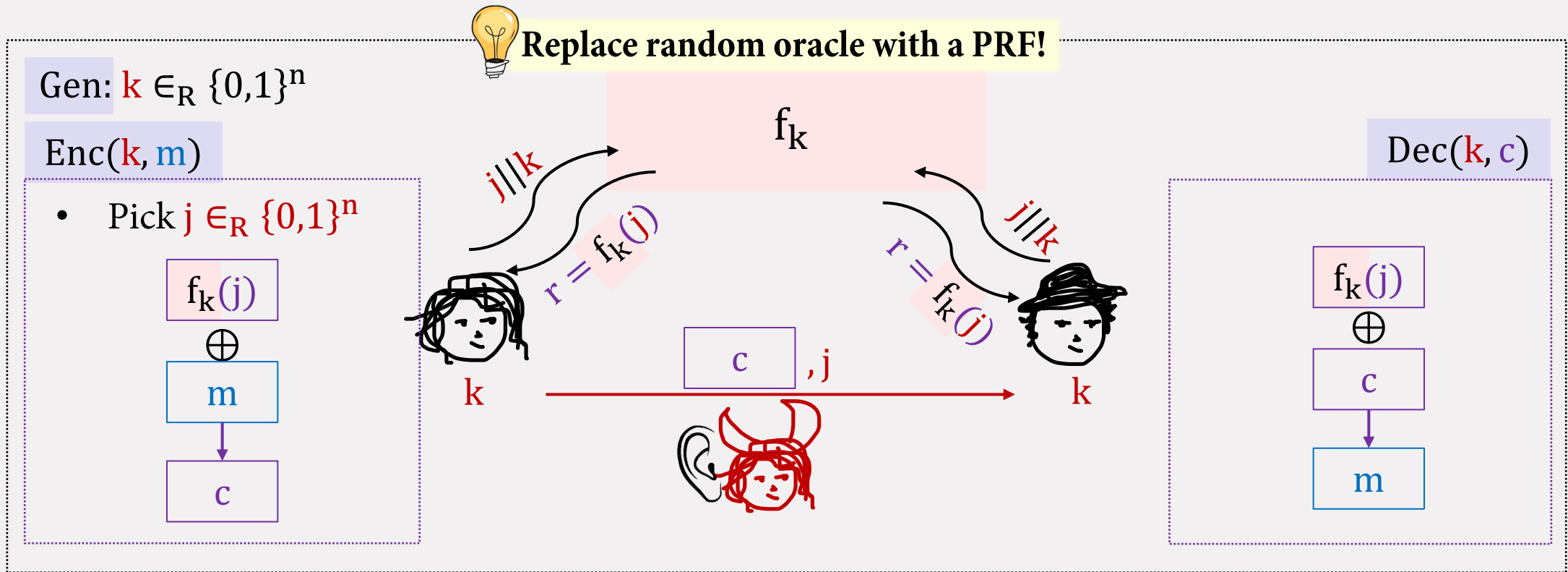


RECALL



Replace random oracle with a PRF!

Stateless Symmetric Key Encryption



Construction 1: Randomized Computational One Time Pad

HW 5: Prove that **Construction 1** is computationally secure for multiple messages against all **PPT** eavesdroppers Eve!

How to build PRFs?

Two Approaches

Theoretical Constructions

Practical Constructions

Rely on well-studied hard problems!

PRGs exist $\Rightarrow$ PRFs exist

[by Goldreich, Goldwasser, and Micali, 1984]

Far from practical!

Start from a design framework
("Appropriately chosen" functions, composed
"appropriately many times" look random)

Build a candidate construction

Do extensive cryptanalysis

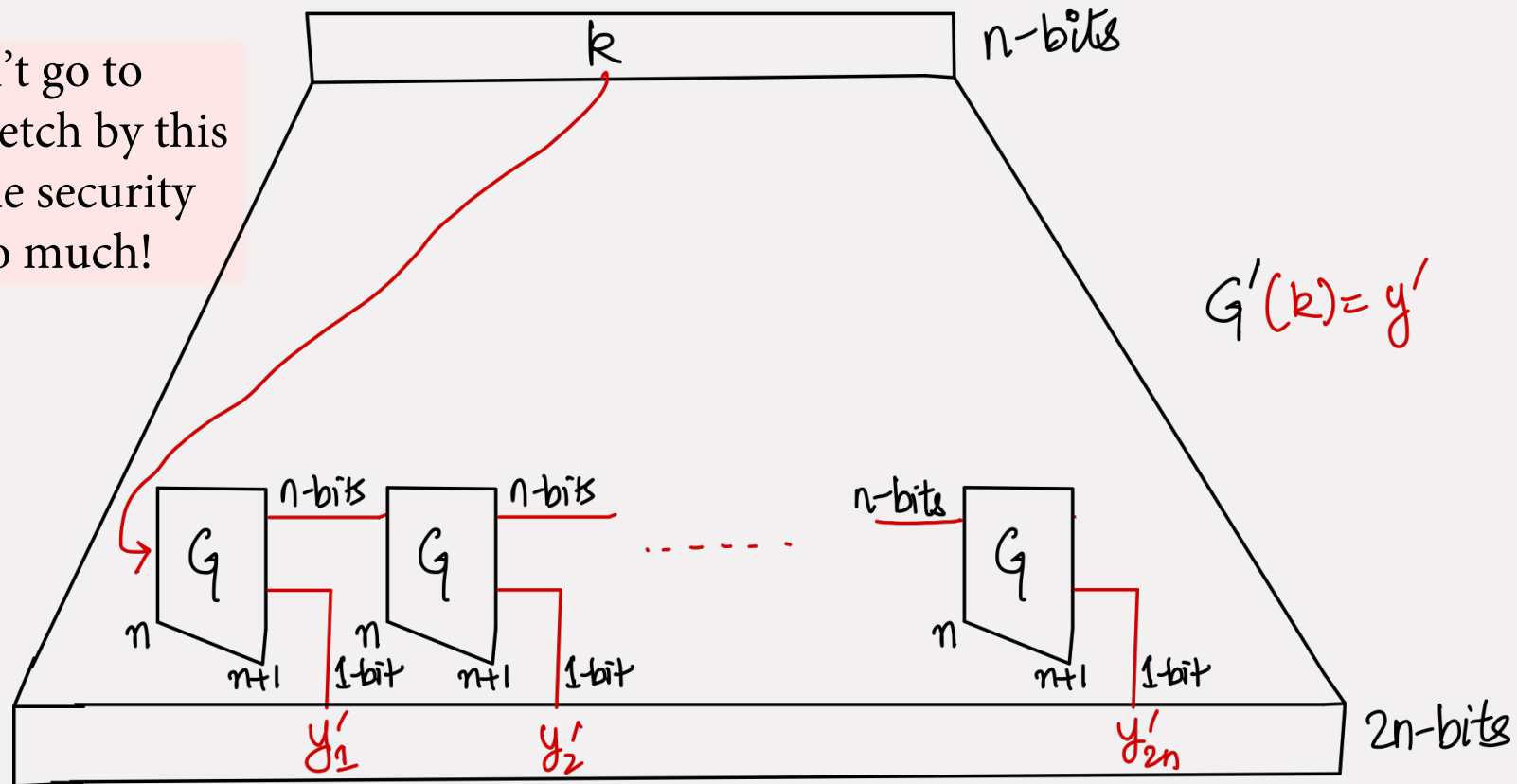
Block Ciphers: AES, DES, etc. [[Boneh and Shoup](#)]

Practical, but not provably secure!

How to build PRFs from PRGs? [GGM]

Recall: The construction of length-extending PRG from last lecture

We couldn't go to exponential stretch by this method, as the security loss was too much!



We need to get to exponential stretch, with efficient accessibility of outputs. How do we do this?

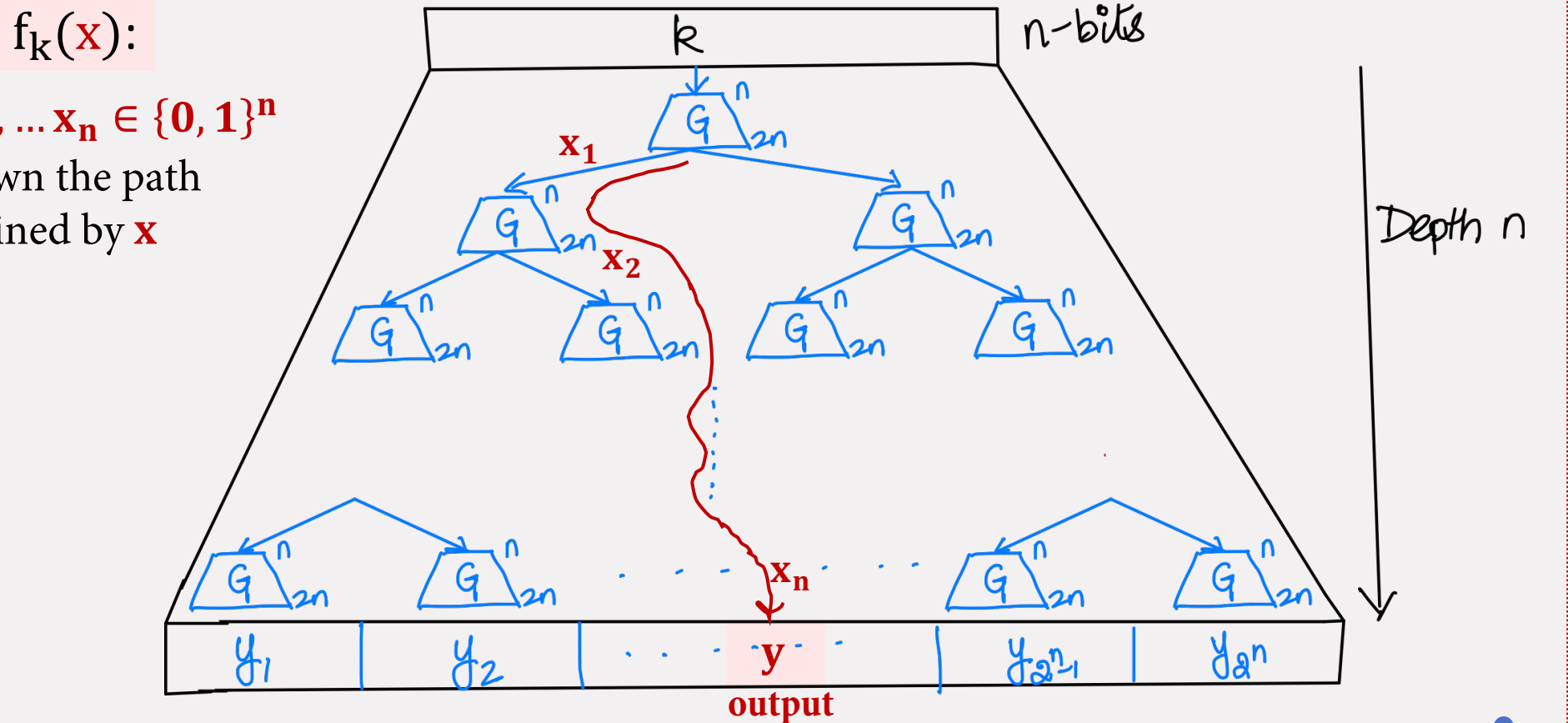
How to build PRFs from PRGs? [GGM]



Use length doubling PRG and use a tree instead of a chain!

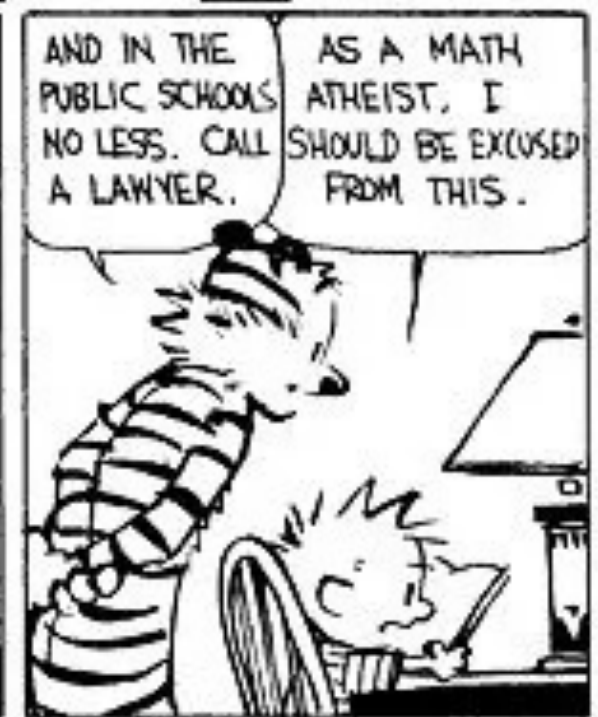
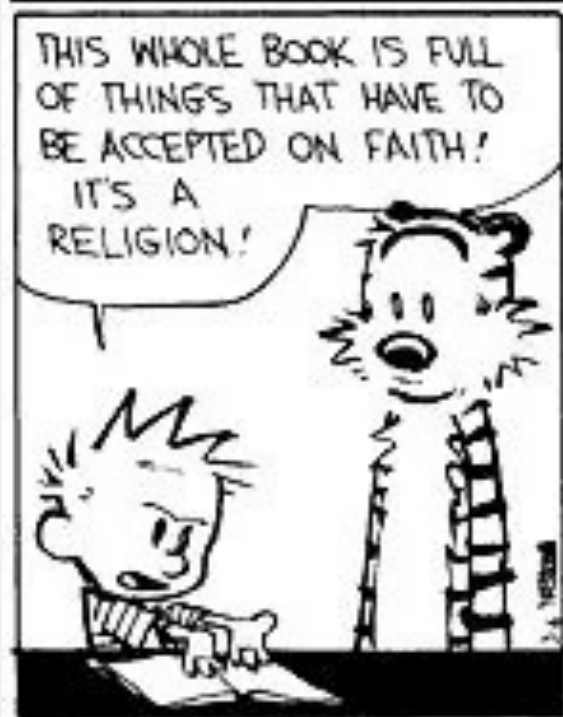
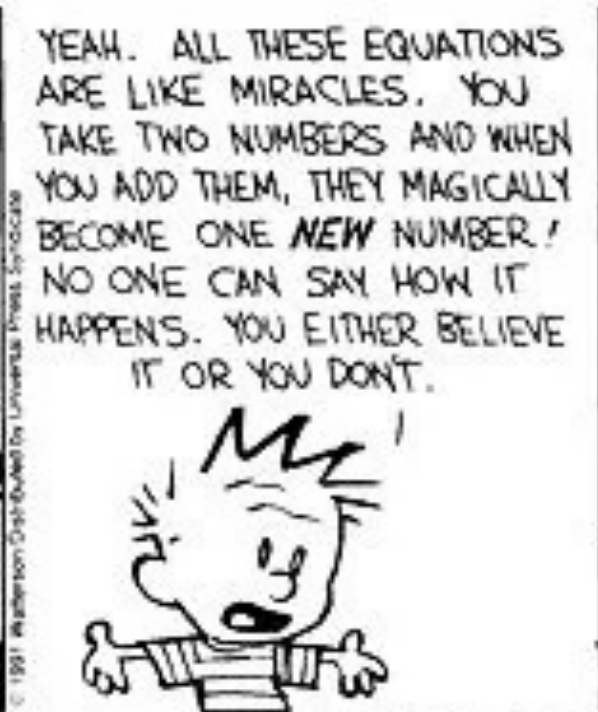
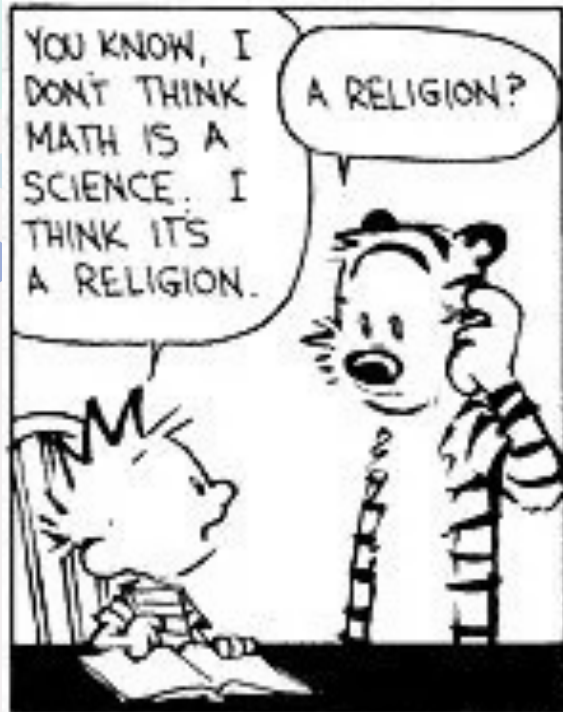
$\mathbf{x} = x_1, x_2, \dots, x_n \in \{0, 1\}^n$

Travel down the path
determined by $\mathbf{x}$



Next Class

- In fact, the symmetric key encryption that we built today is secure against a stronger Eve! (Security against Chosen Plaintext Attack (CPA))
- Block cipher modes of operation (practical ways to encrypt longer messages).



THANK YOU